DATA MANAGEMENT PLAN (DMP)

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Research topic: AI-based multi-objective optimization of desalination systems under renewable energy variability

1. Purpose and scope

This Data Management Plan (DMP) describes the data used in my PhD on AI-assisted decision support for concentrating solar power (CSP)-driven multi-effect distillation with thermal vapour compression (MED-TVC). It covers data origin, formats, quality, storage, access, sharing and preservation.

2. Data description

The principal dataset contains 1,800 steady-state Engineering Equation Solver (EES) simulations. Inputs are exhaust-gas temperature (235–325 °C), mass flow (0.700–0.800 kg/s) and pressure (1.00–1.20 bar). Outputs include distillate production (m^3/day), gain output ratio (GOR), recovery ratio (RR), specific power consumption (kWh/m^3) and pumping power (kW). Another 25 off-grid Latin-hypercube cases are reserved for independent validation. These are numerical simulations, not new plant measurements.

3. Formats, documentation and FAIR

Tabular data will be maintained in CSV (UTF-8) where practical; analysis code in Python (.py/.ipynb); and original models in EES format. A README and data dictionary will record variable definitions, units, parameter ranges, provenance, processing steps, software versions and limitations. FAIR principles will guide documentation and controlled discoverability, interoperability and reuse; FAIR does not require public release.

4. Quality and reproducibility

Raw simulation results, processed tables and independent validation cases will be separated. Checks will address duplicates, missing values, units and operating bounds. Scripts, dependencies, seeds and validation procedures will be documented. The 25 off-grid cases will not be used for training or model selection. Solver convergence is not presented as experimental validation or as proof of case-by-case physical feasibility.

5. Storage, backup and security

My research files are stored on Google Drive and backed up on a USB flash drive. These provide two copies; a third secure copy is planned to better meet the 3-2-1 backup rule. Access will be restricted to authorized collaborators, accounts protected with strong authentication, and backups checked periodically. Confidential data and EES models will not be placed in public repositories.

6. Ownership and confidentiality

My supervisors advise against making the datasets publicly available. Therefore, full datasets, EES models and collaborative code will remain private. Any external transfer or release requires permission from the appropriate supervisors, partners and rights holders. Simulation tables normally contain no personal data; any personal information encountered will be handled under applicable rules, including Moroccan Law 09-08 when relevant.

7. Sharing and licensing

Only authorized project members will access restricted files. Published findings, non-confidential methods and bibliographic metadata can be shared. Depositing datasets or code in Zenodo or another repository would require prior approval; no open license will be applied to collaborative material without authorization.

8. Retention and archiving

Key datasets, documented versions, validation records and authorized scripts will be retained during the PhD. The post-defence retention period and approved long-term archive have not yet been established; these will be confirmed with my university, supervisors and project partners.

9. Risks and review

Main risks are accidental loss, unauthorized access, incomplete documentation and release without permission. Mitigation includes access controls, a planned third backup, regular recovery checks and review of sharing permissions. This living DMP will be updated when datasets, collaborators or institutional requirements change.